

WORK PROCEDURE

PRO-OP-001A

PROCEDURE FOR CHANGING BITS ON THE CUBEX

OBJECTIF

The objective of this procedure is to define a safe way to change bits on the CUBEX

APPLICATION

This procedure should be followed by all MRI employees

INTERPRETATION

The personnel with the authority to interpret this document is the Operations Manager. All requests for amendments, revisions or updates must be made to this person.

RESPONSABILITIES

Supervisors must ensure that workers know and apply this procedure. Workers must know and apply this procedure.

PROCÉDURE

**THE CONSOLE MUST ALWAYS BE MORE THAN 1.5 METERS FROM THE
MAST**

**IT IS FORBIDDEN TO MANOEUVRE THE CONTROLS AT THE SAME TIME
AS THE OPERATOR IS HANDLING THE BIT OR RODS**

**NEVER EXPOSE YOUR HANDS IN THE LINE OF FIRE OF THE BITS OR
RODS**

PROCEDURE FOR CHANGING BITS ON THE CUBEX

STEPS FOR 3 5/8 TO 5 1/8 BITS AND 6 INCHES CASING BITS:

- 1- Ensure a stable floor around the mast



- 2- Position the gear box completely away from the tables



- 3- Make sure the hammer is firmly screwed

- 4- Make sure that the spline piston IS ON



PROCEDURE FOR CHANGING BITS ON THE CUBEX

5- The mast can be to any degree

6- Stop the drill



7- Install the driver sub and retaining rings on the bit



8- Grease the threads of the driver sub well



PROCEDURE FOR CHANGING BITS ON THE CUBEX

9- Position yourself correctly to lift the bit



10-Make sure you have a firm grip on the bit



11-Align the SHANK in the end of the hammer, insert it as far as possible and screw in the bit a minimum of 2 turns.



12-Position the drill at the drilling angle of the hole

13-Open the jaws

PROCEDURE FOR CHANGING BITS ON THE CUBEX

14-Put the bit between the jaws

15-Close the jaws on the bit



16-Tighten the bit with a minimum of 1000 psi and no more than 2000 psi



PROCEDURE FOR CHANGING BITS ON THE CUBEX

STEPS for 6 1/2, 8 and 10 inches

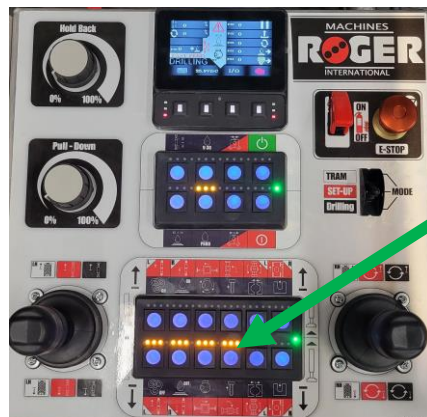
- 1- Ensure you have a stable floor around the mast



- 2- Position the gear box completely away from the tables



- 3- Make sure the hammer is firmly screwed down
- 4- Make sure that the spline piston IS OFF.



PROCEDURE FOR CHANGING BITS ON THE CUBEX

- 5- Swing the mast to zero degree
- 6- Stop the drill



- 7- Install the driver sub and retaining rings on the bit.



- 8- Grease the threads of the driver sub well

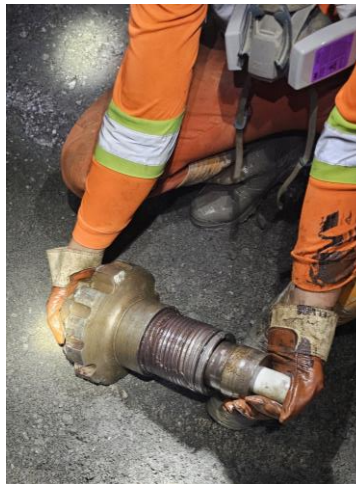


PROCEDURE FOR CHANGING BITS ON THE CUBEX

- 9- Position yourself correctly to lift the bit



- 10- Make sure you have a firm grip on the bit



- 11- Align the SHANK in the end of the hammer and insert it as far as possible and hand screw the bit a minimum of 2 turns.



- 12- Position the drill at the drilling angle of the hole

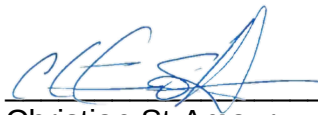
- 12- Open the tables

PROCEDURE FOR CHANGING BITS ON THE CUBEX

- 13- Put the bit between the tables
- 14- Close the tables on the bit



- 15- Tighten the bit with a minimum of 1000 psi and no more than 2000 psi



Christian St-Amour
Operations Manager